

Lecture-5

Mathematics 1 (15B11MA111)

Module: Partial Differentiation

CO [C105.1]

Topics to be covered

➤ Higher Order Partial Derivatives

- ❖ Definition
- ❖ Examples
- ❖ Assignment

➤ Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Higher Order Partial Derivatives

Let $z = f(x,y)$ be a function of two variables and let its first order partial derivatives exist at all points in the domain of definition D of the function f . Then, the first order derivatives are also the functions of x and y . We define the second order partial derivatives as

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left[\frac{\partial f}{\partial x} \right] = f_{xx}(x, y) = \lim_{\Delta x \rightarrow 0} \left[\frac{f_x(x + \Delta x, y) - f_x(x, y)}{\Delta x} \right]$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left[\frac{\partial f}{\partial x} \right] = f_{yx}(x, y) = \lim_{\Delta y \rightarrow 0} \left[\frac{f_x(x, y + \Delta y) - f_x(x, y)}{\Delta y} \right]$$

(differentiate partially first with respect to x and then with respect to y)

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left[\frac{\partial f}{\partial y} \right] = f_{xy}(x, y) = \lim_{\Delta x \rightarrow 0} \left[\frac{f_y(x + \Delta x, y) - f_y(x, y)}{\Delta x} \right]$$

(differentiate partially first with respect to y and then with respect to x)

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left[\frac{\partial f}{\partial y} \right] = f_{yy}(x, y) = \lim_{\Delta y \rightarrow 0} \left[\frac{f_y(x, y + \Delta y) - f_y(x, y)}{\Delta y} \right]$$

If the limit exists. The derivatives f_{xy} and f_{yx} are called mixed derivatives.

➤ Important Remarks:

- ❖ **Remark 1:** If f_{xy} and f_{yx} are continuous at a point $P(x,y)$, then at this point $f_{xy} = f_{yx}$. That is the order of differentiation is immaterial in this case.
- ❖ **Remark 2:** If all the four partial derivatives of second order (f_{xx} , f_{yx} , f_{xy} , f_{yy}) exist at all points in D , then these derivatives are also functions of x and y and can be further differentiated.

Examples

Example 1: Find all the second order partial derivatives of the function

$$f(x, y) = 3x^4 - 2x^3y^2 + 7y$$

Solution: We have

$$f_x(x, y) = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} (3x^4 - 2x^3y^2 + 7y) = 12x^3 - 6x^2y^2$$

$$f_y(x, y) = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (3x^4 - 2x^3y^2 + 7y) = -4x^3y + 7$$

$$f_{xx}(x, y) = \frac{\partial}{\partial x} \left[\frac{\partial f}{\partial x} \right] = \frac{\partial}{\partial x} (12x^3 - 6x^2y^2) = 36x^2 - 12xy^2$$

$$f_{yy}(x, y) = \frac{\partial}{\partial y} \left[\frac{\partial f}{\partial y} \right] = \frac{\partial}{\partial y} (-4x^3y + 7) = -4x^3$$

$$f_{xy}(x, y) = \frac{\partial}{\partial x} \left[\frac{\partial f}{\partial y} \right] = \frac{\partial}{\partial x} (-4x^3y + 7) = -12x^2y$$

$$f_{yx}(x, y) = \frac{\partial}{\partial y} \left[\frac{\partial f}{\partial x} \right] = \frac{\partial}{\partial y} (12x^3 - 6x^2y^2) = -12x^2y$$

Example 2: For the function

$$f(x, y) = \begin{cases} \frac{xy(2x^2 - 3y^2)}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

Show that $f_{xy}(0, 0) \neq f_{yx}(0, 0)$ and also discuss the continuity of f_{xy}, f_{yx} at $(0, 0)$

Solution:

$$f_x(0, 0) = \left. \frac{\partial f}{\partial x} \right|_{(0, 0)} = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0, 0)}{\Delta x} = 0$$

$$f_y(0, 0) = \left. \frac{\partial f}{\partial y} \right|_{(0, 0)} = \lim_{\Delta y \rightarrow 0} \frac{f(0, \Delta y) - f(0, 0)}{\Delta y} = 0$$

$$\begin{aligned}
 f_x(0, y) &= \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, y) - f(0, y)}{\Delta x} \\
 &= \lim_{\Delta x \rightarrow 0} \frac{y[2(\Delta x)^2 - 3y^2]\Delta x}{[(\Delta x)^2 + y^2]\Delta x} = -3y
 \end{aligned}$$

$$\begin{aligned}
 f_y(x, 0) &= \lim_{\Delta y \rightarrow 0} \frac{f(x, \Delta y) - f(x, 0)}{\Delta y} \\
 &= \lim_{\Delta y \rightarrow 0} \frac{x[2x^2 - 3(\Delta y)^2]\Delta y}{[x^2 + (\Delta y)^2]\Delta y} = 2x
 \end{aligned}$$

Now,

$$f_{xy}(0,0) = \frac{\partial}{\partial x} \left[\frac{\partial f}{\partial y} \right] \Big|_{(0,0)} = \lim_{\Delta x \rightarrow 0} \left[\frac{f_y(\Delta x, 0) - f_y(0, 0)}{\Delta x} \right]$$

$$f_{xy}(0,0) = \lim_{\Delta x \rightarrow 0} \frac{2\Delta x - 0}{\Delta x} = 2$$

$$f_{yx}(0,0) = \frac{\partial}{\partial y} \left[\frac{\partial f}{\partial x} \right] \Big|_{(0,0)} = \lim_{\Delta y \rightarrow 0} \left[\frac{f_x(0, \Delta y) - f_x(0, 0)}{\Delta y} \right]$$

$$f_{yx}(0,0) = \lim_{\Delta y \rightarrow 0} \frac{-3\Delta y - 0}{\Delta y} = -3$$

Thus, **$f_{xy}(0, 0) \neq f_{yx}(0, 0)$.**

Since $f_{xy}(0, 0) \neq f_{yx}(0, 0)$, and f_{xy} and f_{yx} are not continuous at $(0, 0)$.

Example 3: For the function

$$f(x, y) = e^{ax} \sin by$$

Show that $f_{xx} + f_{yy} + 2f_{xy} - 2f_{yx} = (a^2 - b^2)e^{ax} \sin by$

Solution: $\frac{\partial f}{\partial x} = ae^{ax} \sin by$

$$\frac{\partial f}{\partial y} = be^{ax} \cos by$$

$$f_{xx}(x, y) = \frac{\partial}{\partial x} \left[\frac{\partial f}{\partial x} \right] = \frac{\partial}{\partial x} (ae^{ax} \sin by) = a^2 e^{ax} \sin by$$

$$f_{yy}(x, y) = \frac{\partial}{\partial y} \left[\frac{\partial f}{\partial y} \right] = \frac{\partial}{\partial y} (be^{ax} \cos by) = -b^2 e^{ax} \sin by$$

$$f_{xy}(x, y) = \frac{\partial}{\partial x} \left[\frac{\partial f}{\partial y} \right] = \frac{\partial}{\partial x} (be^{ax} \cos by) = abe^{ax} \cos by$$

$$f_{yx}(x, y) = \frac{\partial}{\partial y} \left[\frac{\partial f}{\partial x} \right] = \frac{\partial}{\partial y} (ae^{ax} \sin by) = abe^{ax} \cos by$$

Now we can check

$$\begin{aligned} f_{xx} + f_{yy} + 2f_{xy} - 2f_{yx} \\ = a^2 e^{ax} \sin by - b^2 e^{ax} \sin by + 2abe^{ax} \cos by \\ - 2abe^{ax} \cos by = (a^2 - b^2) e^{ax} \sin by \end{aligned}$$

Try Yourself

Q.1: If $f(x, y) = \tan^{-1} \frac{y}{x}$, then find all second order partial derivatives and also verify that $f_{xx} + f_{yy} = 0$.

Q.2: For the function

$$f(x, y) = \begin{cases} \frac{(x^3 y - x y^3)}{x^2 + 2y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

Show that $f_{xy}(0, 0) \neq f_{yx}(0, 0)$

THANK YOU

